

E-11-CX-nabla-xi v1: direction regularity and coercive dissipation audit

ZFL Lab

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MISSION: Test if $\xi = \omega/|\omega|$, $|\nabla\xi|$ provides C_X or W_- control via diffusion coercivity, without e_i frame. Kill fast if no link to W_- divergence.

INPUT: $\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$, E-05 to E-10 BLOCKED/KILLED, G locked. No spectral e_i .

FORBIDDEN: e_i, λ_i, η_- pointwise alignment reuse. $|\nabla\xi| \in L^p \Rightarrow$ regularity without W_- link. Helicity H import. $C_E^{weak} \Rightarrow C_E^{strong}$.

AUDIT:

1. Exact vorticity direction equation:

$\xi = \omega/|\omega|$ where $|\omega| > 0$. Then $\omega = |\omega|\xi$.

Vorticity: $D_t\omega = S\omega + \nu\Delta\omega$.

Take $D_t|\omega| = (\xi^T S\xi)|\omega| + \nu\xi \cdot \Delta\omega$.

But $\Delta\omega = \Delta(|\omega|\xi) = \xi\Delta|\omega| + 2\nabla|\omega| \cdot \nabla\xi + |\omega|\Delta\xi$.

Project orthogonal to ξ : $D_t\xi = S\xi - (\xi^T S\xi)\xi + \nu(\Delta\xi + 2\frac{\nabla|\omega|}{|\omega|} \cdot \nabla\xi + |\nabla\xi|^2\xi)$ plus terms preserving $|\xi| = 1$.

Key identity for enstrophy diffusion:

$\nu \int |\nabla\omega|^2 = \nu \int |\nabla|\omega||^2 + \nu \int |\omega|^2 |\nabla\xi|^2$.

Thus

$$\nu \|\nabla\omega\|_2^2 = \nu \|\nabla|\omega|\|_2^2 + \nu \int |\omega|^2 |\nabla\xi|^2$$

Coercive term: $\nu \int |\omega|^2 |\nabla\xi|^2$ appears in dissipation.

2. Link to W :

$W = \int |\omega|^2 (\xi^T S\xi)$.

No direct $|\nabla\xi|$ in W . But evolution of ξ contains $S\xi - (\xi^T S\xi)\xi$.

If $|\nabla\xi|$ large, vorticity direction highly oscillatory, diffusion large: term $\nu \int |\omega|^2 |\nabla\xi|^2$ consumes enstrophy production. From enstrophy balance:

$\frac{1}{2} \frac{d}{dt} \|\omega\|_2^2 = W - \nu \|\nabla|\omega|\|_2^2 - \nu \int |\omega|^2 |\nabla\xi|^2$.

Thus

$$W = \frac{1}{2} \frac{d}{dt} \|\omega\|_2^2 + \nu \|\nabla|\omega|\|_2^2 + \nu \int |\omega|^2 |\nabla\xi|^2.$$

Compare with original $W = \frac{1}{2} + \nu \|\nabla\omega\|_2^2$, consistent.

Now W_- ? From identity, $W_- \leq$? From identity, $W =$ positive dissipation $+\frac{1}{2}$; so if $\int W \rightarrow \infty$, then $\int \nu |\omega|^2 |\nabla\xi|^2$ may also diverge, but both positive. No sign separation.

Consider: $W_+ - W_- = \frac{1}{2} + D_1 + D_2$ where $D_2 = \nu \int |\omega|^2 |\nabla\xi|^2 \geq 0$.

Thus $W_+ = W_- + \frac{1}{2} + D_1 + D_2$. If W_- diverges in L^1 , W_+ must diverge even more to keep RHS divergence. No contradiction, both may diverge together.

Need lower bound for W_- via $|\nabla\xi|$? Not found.

Thus $|\nabla\xi|$ provides coercive dissipation but not sign separation for W_- .

Potential C_X from CF/DHY: If $\|\nabla\xi\|_\infty$ controlled, then W_- bounded? CF: if direction regular in high vorticity region, no blow-up. This uses $|\nabla\xi|$ bounded.

But no L_t^1 bound from PROVEN.

Output:

$$\nu\|\nabla\omega\|_2^2 = \nu\|\nabla|\omega|\|_2^2 + \nu\int |\omega|^2|\nabla\xi|^2 \text{ EXACT}$$

$$W = \frac{1}{2}\frac{d}{dt}\|\omega\|_2^2 + \nu\|\nabla|\omega|\|_2^2 + \nu\int |\omega|^2|\nabla\xi|^2$$

Coercive $\nu\int |\omega|^2|\nabla\xi|^2 \geq 0$ appears, but W_- not controlled by it

$$C_E^{weak} \Rightarrow \frac{1}{2}\|\omega\|_2^2 + \nu\int \|\nabla|\omega|\|_2^2 + \nu\int |\omega|^2|\nabla\xi|^2 \rightarrow \infty, \text{ no contradiction with } W_- \text{ divergence}$$

Classification: PARTIAL/OPEN — geometric depletion gives dissipation structure, but no $W_- \in L_t^1$ or C_X necessarily divergent beyond PROVEN found in single audit. Needs stronger direction regularity assumption not in PROVEN.

$\mathcal{C}_{PROVEN}$ remains $\{C_E^{weak}, C_{L^3}\}$. Branch not killed fully like E-10, but no signal for W_- separation; mark as OPEN/CANDIDATE requiring extra assumption $|\nabla\xi| \leq K|\omega|$ etc which is not PROVEN.

Next: helicity H third priority, or formal closure of Phase E-CX as BLOCKED.